

Hemang Negandhi

8+ yrs exp in BFSI | Senior Data Engineer @ BNP Paribas | Alteryx | SQL | Power BI | Tableau | Python | AI-Driven Data Engineering
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PROFESSIONAL SUMMARY

BFSI-focused (Banking, Financial Services, and Insurance) Data Engineering and Business Intelligence (BI) professional with Master’s degree in computer science from University of Texas and 8+ years of experience in Data Analytics, ETL/ELT pipeline development, and AI-powered systems. Competent in Python, SQL, Power BI (DAX), Tableau, and Alteryx Designer, with exposure to Machine Learning (ML) and Artificial Intelligence (AI) experience — building Generative AI and Agentic AI solutions using Large Language Models (LLMs), Prompt Engineering, RAG (Retrieval-Augmented Generation), and Lang Chain. Passionate about enabling data-driven decision making, Data Governance in the BFSI sector through robust data workflows, and Analytics.

WORK HIGHLIGHTS

- Financial Data Processing:** Processed 422 GB of BFSI financial data monthly using Alteryx Designer for BNP Paribas, empowering 10+ cross-functional teams with actionable BI insight while maintaining Data Governance and Data Quality standards (C3 project).
- Automated Financial Reconciliation:** Engineered an end-to-end automated recon pipeline between AXIOM and TRITON reporting systems using PowerShell, Alteryx, and Python — saving 900+ hours annually and enabling real-time regulatory reporting to Federal regulators via Power BI.
- REST API Integration & Automation:** Built a dynamic analytics tool using the BambooHR REST API and Power BI, driving Data Quality improvements and HR Process Automation — adopted by 15+ stakeholders across the organization.
- ML Recommendation System:** Developed a scalable Machine Learning (ML) solution using Python, Flask, and Google Cloud Platform (GCP) to process 1.5M+ movie rows for predictive recommendations, applying NLP (stemming, lemmatization) and Feature Engineering.

RECENT AI WORK

- Built an autonomous AI job-search agent in Python using JobSpy, REST APIs, and BeautifulSoup to scrape **50+** listings per run from LinkedIn, indeed, and RemoteOK.
- Designed a multi-stage data pipeline with rule-based hard filtering, Pydantic validation, and URL-based deduplication — eliminating irrelevant results before LLM processing, reducing API token consumption by **~60%**
- Engineered an LLM scoring engine using Groq API (LLaMA 3.3-70B) with structured JSON prompt engineering, and explainable 1–10 relevance scoring against dynamic candidate profile.
- Automated end-to-end data output using Pandas to generate dated, score-sorted csv reports.
- Applied AI agent design patterns including tool-using agent architecture, ReAct-style reasoning, cost-optimised LLM orchestration, and modular separation of scraping, scoring, filtering, and output layers using Python, dotenv, httpx.

SOFTWARE SKILLS

Languages: Python, SQL	Web: HTML, PHP, JavaScript
Database: MySQL, MS SQL, Oracle11g	Business Intelligence: Powerbi, QlikSense, Tableau, SSIS
Frameworks/Libraries: Flask, Apache Spark (PySpark), Pandas, NumPy, NLP (NLTK), Pydantic	IDEs/tools: PyCharm, VS Code
Cloud Platforms: AWS (Redshift, EC2, S3), Google Cloud, Azure	Data Engineering: Alteryx, ETL, SSRS, Warehousing, Analytics
AI / ML: Machine Learning, Generative AI, LLMs (Groq/LLaMA 3.3-70B), Prompt Engineering, RAG, LangChain, Agentic AI	Methodology: CI/CD, Data Governance, Statistical Analysis,

EDUCATION

University of Texas at Arlington, USA	Masters in Computer Science	2019
K. J. Somaiya College of Engineering, India	Bachelors in Information Technology	2015

WORK EXPERIENCE

BNP Paribas, Mumbai, India (Senior Data Engineer)

Jul 2020 - Present

- Worked with **Alteryx** to analyze and design workflows to perform ETL. Hands-on experience in performance tuning, and query optimization in Tableau.
- Created TDEs (Tableau Extract) for various projects and scheduled on Server as per requirements.
- Coordinated with **PARIS, and MADRID team** to meet business requirements as well as ad hoc requests.
- Maintained and enhanced C3 FINANCE — core data source processing 422 GB/month, delivering analytics to 10+ teams; improved data availability and reduced downstream query time by 25%.
- Also led in enhancing the financial data capabilities of the above project. This involved migration of data hierarchy from **FIREF to GRECO_PMA**, which increased the bank's financial data granularity. While FIREF had **12** levels of data, GRECO has **30** levels, which enabled the bank to expand insights by **25%**, leading to better-informed decisions.
- Spearheaded the **Deposit and Trend Analysis** project, saving **786** work hours/year and reducing data errors from **10% to 2.5%**. Also

developed custom measures for financial forecasting and risk analysis, improving prediction accuracy by **25%** and enabling the risk team to make more informed decisions.

- Engineered an automated reconciliation pipeline between AXIOM and TRITON reporting systems using **PowerShell** (scheduled file-detection every 10 mins), Alteryx macros, and embedded **Python** — matching records on a composite key (Date + Series + Version), eliminating manual file tracking.
- Saved **900+** hours of manual work annually by automating the end-to-end recon workflow with real-time email alerts at each stage and built-in error logging, giving users instant visibility into status and failures without manual checks.
- Built a real-time regulatory reporting feed into Power BI from automated recon outputs, enabling users to validate figures and submit accurate, timely reports to Federal regulators in the NAR region.
- Designed version-aware, counter-based file detection logic to independently identify new/updated files on both drives, ensuring recon triggers only on validated matched sets — improving accuracy by **75%** and cutting errors from partial or duplicate file arrivals.

BNY Mellon Pershing, New Jersey, USA (Business Data Analyst)

Sep 2019 - Mar 2020

- Worked with Alteryx to develop WFs to perform **ETL (Data Extraction, Transformation, Loading)**.
- Used SQL to extract data and create ad-hoc reports for data visualization on Tableau Server.
- Conducted rigorous data validation in tableau, ensuring accuracy by meticulously matching and verifying numbers against the database, resulting in a **95%** accuracy rate.
- Actively contributed to the development of tableau workbooks for the EPO and corporate finance department.
- Showcased in-depth knowledge of Star dimensional schema and Snowflakes schema.
- Collaborated with cross-functional teams to gather business requirements and translate them into effective BI solutions, enhancing decision-making efficiency.
- Designed and implemented reusable data models in tableau, ensuring scalability and consistency across multiple business units.

AKA Enterprise Solution, New York, USA (Powerbi Intern)

May 2018 - Jul 2018

- Performed migration of in-house app from content workspace to app workspace in Powerbi.
- Participated in the creation of **dynamic visualizations** to track employee performance.
- Established a robust API integration with Bamboo HR to ensure continuous access to real-time employee data. The tool is now used by more than **15 HR team members**, enhancing team efficiency and productivity.

Infogain India Pvt Ltd, Mumbai, India (Software Engineer)

Aug 2015 – Jul 2017

- Collected data in QlikSense from various sources (Oracle, CSV files, MS SQL, MySQL).
- Created dynamic interactive dashboards in **Tableau, Powerbi and QlikSense**.
- Analyzed underlying data for possible discrepancies, investigated errors, and implemented Incremental Load.
- Performed **ETL** using **SQL Server Integration Services (SSIS), QlikView**.
- Created real-time streaming dashboard with real-time KPIs in Tableau.
- Documenting all ETL/ELT projects and workflows including supporting documentation for troubleshooting.
- Develop Tableau workbooks to perform **YTD, QTD and MTD type analysis**.

CloudFronts Technologies, Mumbai, India (Intern)

Jun 2015 - Jun 2015

- Acquired hands-on experience in **SSIS, SSRS and Powerbi**.
- Demonstrated working with different connectors like Oracle, SQL, Twilio, Zendesk to **Powerbi**.

Algonation, Mumbai, India (Intern)

Dec 2014 - Feb 2015

- Gathered client requirements and created data model for healthcare project in **MySQL**.
- Created website using HTML5, CSS3, PHP and JavaScript (**CodeIgniter framework for PHP**).

ACADEMIC PROJECTS (in Masters)

Movie Suggestion System

- Created the application using **Flask Framework** in **Python 3.6**.
- Used packages as **NumPy, NLTK, and Pandas** for word processing (performed **stemming** and **lemmatization**).
- The application used a database of **1.5 M rows** to provide prediction results of genre. Deployed application on GCP.

Online Advising Simulation

- Developed a client-server **multi-threaded** application for an online student-teacher Advising system.
- Implemented message queuing using a producer-consumer design pattern.